

PAPER_893: SCm Phonon Modulated Energy Φ

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Abstract

Phonon-modulated energy $E_{\text{net}}^{\text{phonon}}(t) = E_{\text{net}}(t) \times \Phi_{\{1.25\text{THz}\}}(\omega)$ where $\Phi = \Phi_0 \cdot \exp[-(\omega - \omega_{\text{SCm}})^2 / (2\Gamma^2)] \cdot S_{26}$. Gaussian modulation peaks at the SCm resonance frequency, amplifying the net energy by the phonon flux density.

1. Core Equations

$$\Phi_{\{1.25\text{THz}\}} = \Phi_0 \cdot \exp[-(\omega - \omega_{\text{SCm}})^2 / (2\Gamma^2)] \cdot S_{26}$$
$$E_{\text{net}}^{\text{phonon}} = E_{\text{net}} \times \Phi$$

2. Parameters

Parameter	Default	Description
omega	$2\pi \times 1.25\text{e}12$ rad/s	Angular frequency
E_net_bare	1.0 J	Bare net energy

3. UQFF Integration

This calculator integrates `et_scm_vacuum.py` from Session 207 into the CP4 pipeline as class #477.

PAPER_893 — Star Magic UQFF Framework — CVW v2.0.0